Economics 351X Test 2 Fall 2026 Prof. Hill

Test Version A

Instructions: Please fill out your name on this page and sign the honesty pledge. On the Akindi form, fill in your **name**, **student ID**, and **exam version**. Once you've worked out your answers, transfer them to the Akindi answer sheet. You have 80 min – use your time accordingly.

Name: _____

I pledge that I have neither given nor received aid on this examination.

Signature: _____

Chapter 2

Derivative Rules

General Function	Derivative	Examples
$f(x) = c + ax^b$	$\frac{df(x)}{dx} = abx^{(b-1)}$	$f(x) = 3 + 5x^2 \Rightarrow \frac{df(x)}{dx} = 10x$
$f(x_1, x_2) = ax_1^b + cx_2^d$	$\frac{\partial f(x_1,x_2)}{\partial x_1} = abx_1^{(b-1)}$ $\frac{\partial f(x_1,x_2)}{\partial x_2} = cdx_2^{(d-1)}$	$f(x) = 5x_1^2 + 3x_2 \Rightarrow \frac{\partial f(x_1,x_2)}{\partial x_1} = 10x_1$ $f(x) = 5x_1^2 + 3x_2 \Rightarrow \frac{\partial f(x_1,x_2)}{\partial x_2} = 3$
$f(x_1, x_2) = ax_1^b x_2^c$	$\frac{\partial f(x_1,x_2)}{\partial x_1} = abx_1^{(b-1)}x_2^c$ $\frac{\partial f(x_1,x_2)}{\partial x_2} = acx_1^b x_2^{(c-1)}$	$f(x) = 5x_1^2x_2 \Rightarrow \frac{\partial f(x_1,x_2)}{\partial x_1} = 10x_1x_2$ $f(x) = 5x_1^2x_2 \Rightarrow \frac{\partial f(x_1,x_2)}{\partial x_2} = 5x_1^2$
Chain rule	$h(x) = f(g(x))$ $h'(x) = f'(g(x))g'(x)$	$h(x) = (7 + 3x^2)^5$ $h'(x) = 5(7 + 3x^2)^4 \times (6x)$
Product rule	$h(x) = f(x)g(x)$ $h'(x)=f'(x)g(x)+f(x)g'(x)$	$h(x)=(7 + 3x^2)(6 + 5x^3)$ $h'(x)=(6x)(6+5x^3)+(7+3x^2)(15x^2)$

Chapter 3

Cost

$$C(q) = FC + VC(q).$$

Average Cost, Average Fixed Cost, Average Marginal Cost, Marginal Cost

$AC = \frac{C}{q}.$ $AVC = \frac{VC}{q}.$ $AFC = \frac{FC}{q}$, $MC = \frac{dC}{dq}.$

Revenue=pQ

Profit

$$\text{Profit}(q) = \text{Revenue}(q) - \text{Cost}(q). \qquad \text{Profit}(q) = \text{Variable Profit}(q) - FC,$$

$$\text{Profit}(q) = [p - AC(q)]\, q.$$

Chapter 4

Price Elasticity of Demand

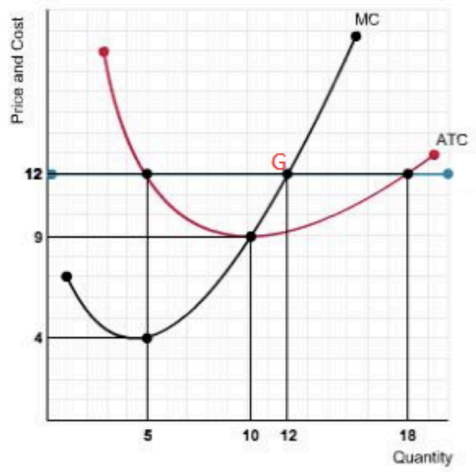
$$E_{Q_1^*,p_1} = \left(\frac{\partial Q_1^*(p_1, p_2, I)}{\partial p_1}\right) \frac{p_1}{Q_1^*(p_1, p_2, I)}.$$

Elasticity= percent change in Q/ percent change in P

Chapter 6

Lerner Index of Market Power

$$L = \frac{p - MC}{p}. = \frac{-1}{E_{Q,p}}.$$



1. Use the above graph to answer the following question. Assume the market is in perfect competition. If the market price is $4, how much will the firm produce in the long run?

a) 5 b) 9 **c) 10** d) 12 e) 0

E also accepted

Use the below information for the next 2 questions.

2. Jon's bike shop operates in a perfectly competitive market where the prevailing price for a bike is $18. If Jon's marginal cost is 3q. Total cost= $1.5q^2+36$.

Profit is _____and the firm _____ in the short run.

a) Positive, continue to operate
b) Positive, shutdown
c) Zero, shutdown
d) Negative, continue to operate
e) Negative, shutdown

3. What will be the long run price in this market?

a) 12
b) 36
c) 18
d) 6
e) None of the above

4. A firm operates in a perfectly competitive market where the prevailing price for their product is $30. The marginal cost function is given by MC=0.5q. If fixed costs are $900, in the short run the firm will (MARK ALL THAT APPLY)

a) Continue to operate
b) Make zero profit
c) Produce 60 units
d) Produce 0 units
e) Have positive variable profit

5. There are currently 10 perfectly competitive firms producing output q. Their cost function is $C=30+3q^2$. The demand in this market is $Q_D=1200-40p$. Total producer surplus is?

a) 48
b) 2880
c) 24
d) 240
e) none of the above

6. A firm is a monopolist, the market demand curve is given by the equation: Q=800-5P, for what range of output will the firm's revenue be increasing?

a) 0<Q<200
b) 0<Q<400
c) Q>600
d) Q>800
e) None of the above

7. Jimmy Johnson's fruit stand operates in a perfectly competitive market where the prevailing price for a coconut (his only product) is $3. Suppose Jimmy's marginal cost function is given by MC=0.1q. For which level of fixed costs should he operate in the short run? (MARK ALL THAT APPLY)

a) FC=0
b) FC=10
c) FC=50
d) FC= 100
e) FC=1000

8. Two firms compete in Cournot duopoly. Market inverse demand curve is given by:

P = 120 - Q where Q = q1 + q2

Each firm has a constant marginal cost: MC = 20 and no fixed costs. Assuming each firm chooses output simultaneously, what is the equilibrium output for each firm?

a) 20 b) 30 c) 40 d) 50 **e) none of the above**

9. What tactic(s) could a firm use to gain market power? (MARK ALL THAT APPLY)

- **a) Regulatory capture**
- **b) Vertical integration**
- **c) Horizontal integration**
- **d) A price war**
- **e) A patent**

10. The demand curve is given by the equation Q=30-2P, the supply curve is given by the equation Q=P. What is producer surplus at the equilibrium price?

- **a) 50**
- b) 75
- c) 100
- d) 125
- e) None of the above

11. A market has the following demand curve P=250-5Q. Two firms operate in Bertrand competition. Each has MC=150 and no fixed costs. What will the TOTAL quantity supplied in the market be?

a) **20** b) 40 c) 50 d) 60 e) None of the above

12. Which of the following markets would likely have the highest Lerner index?

- a) A firm operating in perfect competition
- b) A firm in monopolistic competition with many substitutes
- **c) A firm in monopolistic competition with few substitutes**
- d) A firm in a commoditized market
- e) Bertrand competition

13. Which of the following is **NOT** a demand curve shifter? (MARK ALL THAT APPLY)

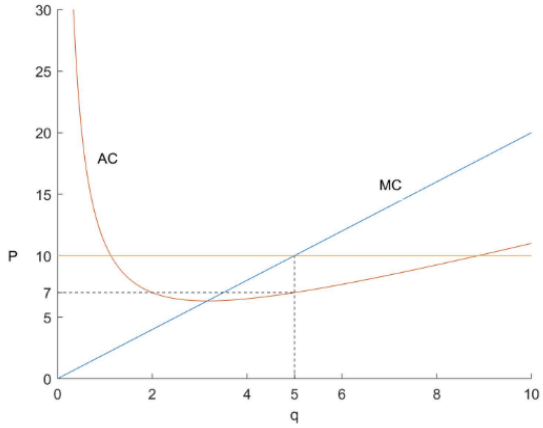
a) Price of the good itself
b) Price of related goods
c) Input prices to make the good
d) Tastes
e) Income

14. The goal(s) of advertising is(are)? (MARK ALL THAT APPLY)

a) To move the demand curve to the right
b) To move the demand curve to the left
c) To make demand more elastic
d) To make demand more inelastic
e) Shift the supply curve to the left

15. Peanut butter becomes cheaper. At the same time, strawberries (an input into jam) become less expensive. What do you expect to happen to the price and quantity sold of jam (a complement to peanut butter) in the jam Market?

a) Price increases, quantity change is indeterminate
b) Price change is indeterminate, quantity increases
c) Price change is indeterminate, quantity decreases
d) Price decreases, quantity decreases
e) Price decreases, quantity change is indeterminate



16. Use the above graph to answer the following question. A firm has the cost curves above. What is its fixed cost?

a) 0
b) 7
c) 5
d) 10
e) None of the above

17. The demand for apples has an elasticity of demand of -0.5. Which of the following would decrease revenue in the market for apples? (MARK ALL THAT APPLY)

a) A price ceiling below the market equilibrium
b) A price floor above the market equilibrium
c) A decrease in price of oranges (a substitute for apples)
d) An increase in price of water (an input into apple production)
e) A new technology that enables cheaper production of apples

18. A firm is engaged in monopolistic competition. Which of the following goods would have the highest mark up?

a) A good with elasticity of -.01
b) A good with elasticity of -.10
c) A good with elasticity of -1
d) A good with elasticity of .1
e) Need more information

19. In the podcast on monopolistic competition, what was the main strategic problem with Stephon Marbury's Starbury sneaker?

a) It was too expensive for its target market
b) It failed because the shoe quality was objectively poor
c) It was "stuck in the middle," trying to be both high-status and low-cost
d) It lacked any celebrity endorsement
e) all of the above



20. The above photo was used in lecture to demonstrate the importance of: (pick the best answer, the best answer is not E)

a) Sunk Cost Fallacy
b) Markets
c) Elasticity
d) Mimetic Desire
e) Doing the practice questions

Use the following information for the next <u>3 questions</u>:

A monopolist faces the following linear inverse demand curve: P=280-4Q where (P) is the price and (Q) is the quantity produced. The monopolist's total cost is: $TC=300+40Q+2Q^2$

21. What price will the monopolist charge?

a) 120
b) 160
c) 180
d) 200
e) None of the above

22. What will be the deadweight loss due to monopoly? You can assume the firm's marginal cost curve is the supply curve under perfect competition.

a) 0
b) 200
c) 400
d) 800
e) None of the above

23. What is the monopolist's Lerner index at its profit-maximizing quantity?

a) 0.20
b) 0.40
c) 0.60
d) 0.67
e) None of the above

24. Universities agreeing to offer similar aid packages would be what type of market structure?

a) Perfect Competition
b) Monopoly
c) Monopolistic Competition
d) Bertrand
e) Colluding Oligopoly

25. There are 2 identical firms engaged in Bertrand competition, each with cost function: C(q)=100+q and Market demand is Q=60-P. What is the deadweight loss at the equilibrium?

a) 1500
b) 1250
c) 250
d) 150
e) 0

26. What is a possible second order effect of price ceilings (MARK ALL THAT APPLY)?

a) The decline in quality of a good
b) Underground markets
c) Discrimination
d) Bribery
e) Long lines

BONUS QUESTIONS (extra credit)

27. According to the video on how to quit, what are some reasons people have a hard time quitting? (MARK ALL THAT APPLY)

a) Identity
b) Status Quo Bias
c) Sunk Cost Fallacy
d) Kill Criteria

28. In the video on the Monopoly board game, what protest do I attend?

a) A protest against war
b) A protest against building condos
c) A protest against tariffs
d) A protest against monopolies
or
e) None of the above